

SINGLE-PHASE GAS FLOW IN MICROCHANNELS

2.1. Rarefaction and wall effects in microflows

- 2.1.1. Gas at the molecular level
 - 2.1.1.1. Microscopic length scales
 - 2.1.1.2. Binary intermolecular collisions in dilute simple gases
- 2.1.2. Continuum assumption and thermodynamic equilibrium
- 2.1.3. Rarefaction and Knudsen analogy
- 2.1.4. Wall effects

2.2. Gas flow regimes in microchannels

- 2.2.1. Ideal gas model
- 2.2.2. Continuum flow regime
 - 2.2.2.1. Compressible Navier–Stokes equations
 - 2.2.2.2. Classic boundary conditions
- 2.2.3. Slip flow regime
 - 2.2.3.1. Continuum NS–QGD–QHD equations
 - 2.2.3.2. First-order slip boundary conditions
 - 2.2.3.3. Higher-order slip boundary conditions
 - 2.2.3.4. Accommodation coefficients
- 2.2.4. Transition flow and free molecular flow
 - 2.2.4.1. Burnett equations
 - 2.2.4.2. DSMC method
 - 2.2.4.3. Lattice Boltzmann method

2.3. Pressure-driven steady slip flows in microchannels

- 2.3.1. Plane flow between parallel plates
 - 2.3.1.1. First-order solution
 - 2.3.1.2. Second-order solutions
- 2.3.2. Gas flow in circular microtubes
 - 2.3.2.1. First-order solution
 - 2.3.2.2. Second-order solution
- 2.3.3. Gas flow in annular ducts
- 2.3.4. Gas flow in rectangular microchannels
 - 2.3.4.1. First-order solution
 - 2.3.4.2. Second-order solution
- 2.3.5. Experimental data
 - 2.3.5.1. Experimental setups for flow rate measurements
 - 2.3.5.2. Flow rate data
 - 2.3.5.3. Pressure data
 - 2.3.5.4. Flow visualization
- 2.3.6. Entrance effects

2.4. Pulsed gas flows in microchannels

2.5. Thermally driven gas microflows and vacuum generation

- 2.5.1. Transpiration pumping
- 2.5.2. Accommodation pumping

2.6. Heat transfer in microchannels

- 2.6.1. Heat transfer in a plane microchannel
 - 2.6.1.1. Heat transfer for a fully developed incompressible flow
 - 2.6.1.2. Heat transfer for a developing compressible flow
- 2.6.2. Heat transfer in a circular microtube
- 2.6.3. Heat transfer in a rectangular microchannel

2.7. Future research needs

2.8. Solved examples

- Example 2.1
- Example 2.2

2.9. Problems

- Problem 2.1
- Problem 2.2
- Problem 2.3
- Problem 2.4
- Problem 2.5
- Problem 2.6

Appendix

Chapter 2-3

2.3. Pressure-driven steady slip flows in microchannels

2.3.1. Plane flow between parallel plates

2.3.1.1. First-order solution

2.3.1.2. Second-order solutions

2.3.2. Gas flow in circular microtubes

2.3.2.1. First-order solution

2.3.2.2. Second-order solution

2.3.3. Gas flow in annular ducts

2.3.4. Gas Flow in rectangular microchannels

2.3.4.1. First-order solution

2.3.4.2. Second-order solution

2.3.5. Experimental data

2.3.5.1. Experimental setups for flow rate measurements

2.3.5.2. Flow rate data

2.3.5.3. Pressure data

2.3.5.4. Flow visualization

2.3.6. Entrance effects

2.3. Pressure-driven steady slip flows in microchannels

In this section, we consider the flow of an ideal gas through different microchannels in the slip-flow regime.

The flow is steady, isothermal and the volume forces (body forces - LK) are neglected.

The flow is also assumed to be locally fully developed:

the velocity profile in a section is the same as the one obtained for a fully developed incompressible flow,

but the density is recalculated in each cross-section and depends on the pressure and the temperature via the equation of state (2.24).

This assumption is no longer valid if the Mach number is greater than 0.3 (Harley *et al.*, 1995; Guo and Wu, 1998).

The main sections of interest are the rectangular and the circular cross-sections.

Microchannels with rectangular cross-sections are easily etched in silicon wafers, for example by reactive ion etching (RIE) or by deep reactive ion etching (DRIE).

If the aspect ratio of the microchannel is small enough, the flow may be considered as a plane flow between parallel plates and the modeling is much simpler.

Gas flows in circular microtubes are also frequently encountered, for example in chromatography applications. Fused silica microtubes are proposed by different suppliers with a variety of internal diameters, from one micrometer to hundreds of micrometers. Due to their low cost, they are an interesting solution as connecting lines for a number of microfluidics devices or prototypes.

The axial coordinate is z and the cross-section is in the (x, y) plane. The different sections considered in this section are represented in Fig. 2.10.

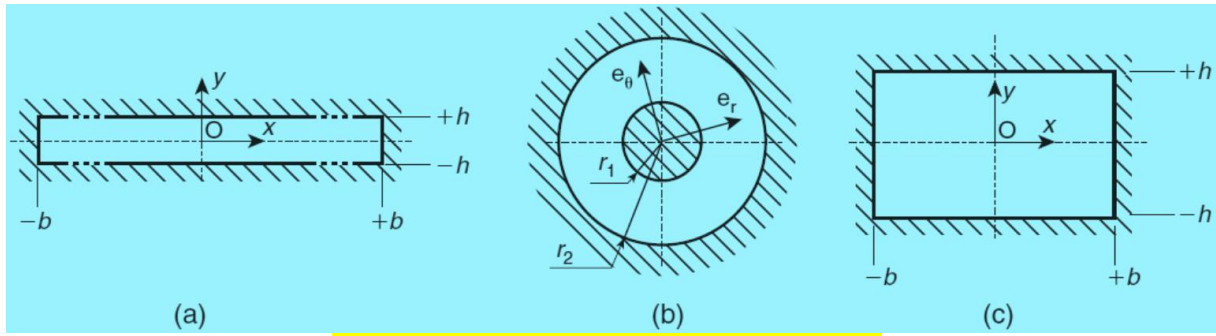


Fig. 2.10. Different microchannel sections.

(a) Plane channel limited by parallel plates ($b \gg h$); (b) circular or annular duct; (c) rectangular channel.

Two different Knudsen numbers are usually defined for microchannel flows.

$$\text{Kn} = \frac{\lambda}{D_H} = \frac{\lambda}{4A/P} \quad (2.84)$$

is defined from the hydraulic diameter as a reference length. The section area is noted A and the wetted perimeter is noted P .

$$\text{Kn}' = \frac{\lambda}{L_{\min}} \quad (2.85)$$

is defined from the minimal representative length of the section.

2.3.1. Plane flow between parallel plates

Here, $h/b \rightarrow 0$ and the velocity only depend on the transverse coordinate y . With the previous assumptions, the momentum equation (2.33) reduces to

$$\frac{d^2 u_z}{dy^2} = \frac{1}{\mu} \frac{dp}{dz} \quad (2.86)$$

(General explicit forms of momentum equations are added to the end of this chapter - LK)

2.3.1.1. First-order solution

The first-order boundary condition (2.58) for a **fixed wall** and an **isothermal flow** is written $u_z|_w = [(2 - \sigma)/\sigma] \lambda \partial u_z / \partial n|_w$, that is for a flow between parallel plates:

$$u_z|_{y=h} = -\frac{2 - \sigma}{\sigma} \lambda \frac{\partial u_z}{\partial y} \Big|_{y=h} \quad (2.87)$$

to be associated with the condition of symmetry (*cf.* Fig. 2.10)

$$\frac{du_z}{dy} \Big|_{y=0} = 0 \quad (2.88)$$

The general solution of Eq. (2.86) is

$$u_z = \frac{1}{\mu} \frac{dp}{dz} \left(\frac{y^2}{2} + a_1 y + a_2 \right) \quad (2.89)$$

The condition of **symmetry** (2.88) yields $a_1 = 0$ and the boundary condition (2.87) leads to $a_2 = -\lambda h \frac{2 - \sigma}{\sigma} - \frac{h^2}{2}$

Therefore, in a **dimensionless** form, the velocity distribution can be written as

$$u_z^* = 1 - y^{*2} + 8 \frac{2 - \sigma}{\sigma} \text{Kn} = 1 - y^{*2} + 4 \frac{2 - \sigma}{\sigma} \text{Kn}' \quad (2.90)$$

where $y^* = y/h$, $u_z^* = u_z / u_{z0}$ and $u_{z0} = u_z(y=0, \text{Kn}=0)$ represents the velocity at the center of the microchannel when rarefaction is not taken into account:

$$u_{z0} = -\frac{h^2}{2\mu} \frac{dp}{dz} \quad (2.91)$$

The Knudsen number $Kn=\lambda/L=\lambda/D_H$ in Eq. (2.90) is defined from the hydraulic diameter $D_H=4A/P=4h$, noting $A=4hb$ as the area and $P=4(b+h)$ as the wetted perimeter of the channel cross-section, with $b \gg h$ (Fig. 2.10(a)).

$Kn'=\lambda/2h$ is defined from the depth of the microchannel.

Equation (2.90) shows that the usual Poiseuille parabolic profile ($u_z^*=1-y^{*2}$) is just globally translated with a quantity $2(2-\sigma)\lambda/(\sigma h)$, due to the slip at the wall.

The integration of the velocity distribution given by Eq. (2.90) over the cross-section leads to the mean velocity

$$\overline{u_z^*} = \frac{1}{2} \int_{-1}^1 u_z^* dy^* \quad (2.92)$$

that is with Eq. (2.90):

$$\overline{u_z^*} = \frac{\overline{u_z}}{u_{z0}} = \frac{2}{3} + 8 \frac{2-\sigma}{\sigma} Kn = \frac{2}{3} + 4 \frac{2-\sigma}{\sigma} Kn' \quad (2.93)$$

We can now calculate the Poiseuille number Po , defined as:

$$Po = C_f Re \quad (2.94)$$

where

$$C_f = \frac{\overline{\tau_w}}{\frac{1}{2} \rho \overline{u_z}^2} \quad (2.95)$$

is the friction factor that represents a dimensionless value of the average wall shear $\overline{\tau_w}$ and

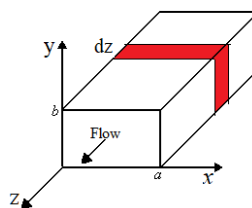
$$Re = \frac{\rho \overline{u_z} D_H}{\mu} \quad (2.96)$$

is the Reynolds number.

The average wall shear is obtained from the force balance

$$\overline{\tau_w} P dz = -A dp \quad (2.97)$$

on a control volume that spans the channel and has an axial extension dz .



Therefore,

$$\overline{\tau_w} = -\frac{D_H}{4} \frac{dp}{dz} \quad (2.98)$$

and

$$Po = -\frac{D_H^2}{2\mu} \frac{dp}{dz} \frac{1}{\overline{u_z}} \quad (2.99)$$

which leads with Eqs. (2.91) and (2.93) to:

$$Po_{NSI,plan} = \frac{24}{1 + 12 \frac{2-\sigma}{\sigma} Kn} = \frac{24}{1 + 6 \frac{2-\sigma}{\sigma} Kn'} \quad (2.100)$$

Equation (2.100) shows that the **Poiseuille number** is **less than** its **usual value of 24** for a Poiseuille flow between parallel plates (Shah, 1975), as soon as the **Knudsen number** is no longer **negligible**.

This result (**Po < 24 LK**) shows that **slip** at the wall **logically reduces friction** and consequently, for a given **pressure gradient**, the **flow rate is increased**.

Considering a long microchannel of length l with a **pressure p_i at the inlet** and a **pressure p_o at the outlet** and neglecting the entrance effects, we can deduce from Eq. (2.93) the mass flow rate $\dot{m} = \rho \overline{u_z} A = p \overline{u_z} A / RT$, which is independent of z . According to Eq. (2.30),

$$Kn = \frac{k_2 \mu(T)}{\rho \sqrt{RT} L} = \frac{k_2 \mu(T) \sqrt{RT}}{pL} \quad (2.30)$$

the **quantity (Kn p)** is **constant** for an **isothermal** flow and it can be written as a function of the outlet conditions: $Kn p = Kn_o p_o$. Therefore, the integration of Eq. (2.93) yields

$$\begin{aligned}
\dot{m} &= \rho \bar{u}_z A = \frac{p \bar{u}_z A}{RT} = \frac{p \bar{u}_z^* u_{z0} A}{RT} = -p \bar{u}_z^* \frac{A}{RT} \frac{h^2}{2\mu} \frac{dp}{dz} \\
\dot{m} \frac{RT}{A} \frac{2\mu}{h^2} dz &= p \bar{u}_z^* dp \quad \dot{m} \frac{RT}{A} \frac{2\mu}{h^2} \int_0^l dz = \int_{p_i}^{p_o} p \bar{u}_z^* dp \\
\int_{p_i}^{p_o} p \bar{u}_z^* dp &= \int_{p_i}^{p_o} p \left(\frac{2}{3} + 8 \frac{2-\sigma}{\sigma} \text{Kn} \right) dp = \int_{p_i}^{p_o} \left(\frac{2}{3} p + 8 \frac{2-\sigma}{\sigma} \text{Kn} p \right) dp = \\
\int_{p_i}^{p_o} \left(\frac{2}{3} p + 8 \frac{2-\sigma}{\sigma} \text{Kn}_o p_o \right) dp &= -\frac{2\mu \dot{m} RT l}{h^2 A} \quad (2.101)
\end{aligned}$$

$$\frac{\bar{u}_z^*}{u_{z0}} = \frac{\bar{u}_z}{u_{z0}} = \frac{2}{3} + 8 \frac{2-\sigma}{\sigma} \text{Kn} = \frac{2}{3} + 4 \frac{2-\sigma}{\sigma} \text{Kn}' \quad (2.93)$$

which leads to

$$\dot{m}_{\text{NS1,plan}} = \frac{2bh^3 p_o^2}{\mu RT l} \left[\frac{\Pi^2 - 1}{3} + 8 \frac{2-\sigma}{\sigma} \text{Kn}_o (\Pi - 1) \right] \quad (2.102)$$

where $\Pi = p_i/p_o$ is the ratio of the inlet over outlet pressures and Kn_o is the outlet Knudsen number. The subscript **NS1** refers to the **Navier–Stokes equations with first-order** slip flow boundary conditions.

In dimensionless form that compares the actual mass flow rate $\dot{m}$ with the mass flow rate $\dot{m}_{\text{ns}}$ obtained from a no-slip hypothesis, Eq. (2.102) becomes:

$$\dot{m}_{\text{NS1,plan}}^* = \frac{\dot{m}_{\text{NS1,plan}}}{\dot{m}_{\text{ns,plan}}} = 1 + 24 \frac{2-\sigma}{\sigma} \frac{\text{Kn}_o}{\Pi + 1} = 1 + 12 \frac{2-\sigma}{\sigma} \frac{\text{Kn}_o'}{\Pi + 1} \quad (2.103)$$

The **pressure distribution** along the z-axis can be found by expressing the **conservation of mass flow rate** through the channel:

$$\Pi = (p_i/p_o) = p/p_o$$

$$\begin{aligned} \dot{m}_{NS1,plan} &= \frac{2bh^3 p_o^2}{\mu RTl} \left[\frac{\Pi^2 - 1}{3} + 8 \frac{2-\sigma}{\sigma} Kn_o (\Pi - 1) \right] \\ &= \frac{2bh^3 p_o^2}{\mu RT(l-z)} \left[\frac{(p/p_o)^2 - 1}{3} + 8 \frac{2-\sigma}{\sigma} Kn_o (p/p_o - 1) \right] \end{aligned} \quad (2.104)$$

Equation (2.104) shows that the pressure $p^* = p/p_o$ is the solution of the polynomial

$$(\text{Right side of (2.104)} - \dot{m}_{NS1,plan} = 0 \quad \text{LK}) \Rightarrow$$

$$p^{*2} + \alpha_1 p^* + \alpha_2 + \alpha_3 z^* = 0 \quad (2.105)$$

where $z^* = z/l$ and

$$\alpha_1 = 24 \frac{2-\sigma}{\sigma} Kn_o \quad \alpha_2 = -\Pi(\Pi + \alpha_1) \quad \alpha_3 = (\Pi - 1)(\Pi + 1 + \alpha_1) \quad (2.106)$$

Equation (2.105) is plotted in **Fig. 2.11**, for an:

inlet over outlet pressure ratio $\Pi = 2$.

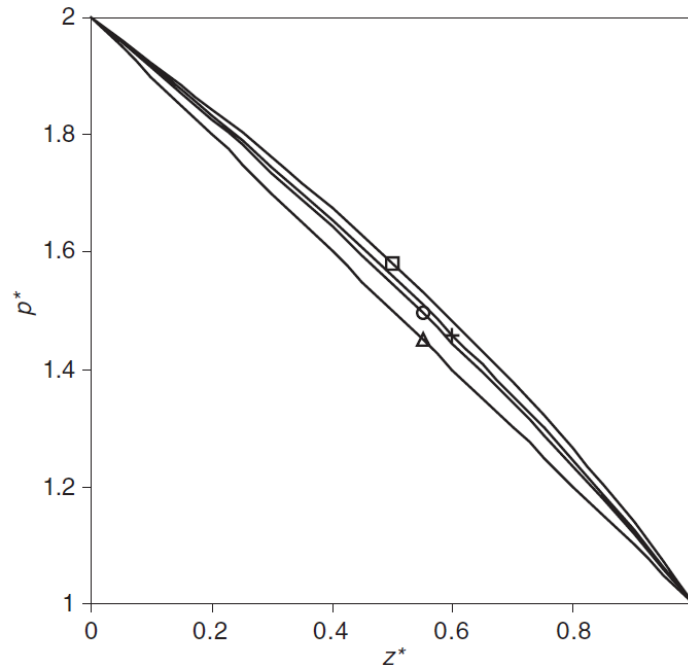


Fig. 2.11. Pressure distribution along a plane microchannel.

Δ: incompressible flow; □: $Kn_o = 0$; +: $Kn_o = 0.05$; ○: $Kn_o = 0.1$.

This figure shows that rarefaction and compressibility have opposite effects on the pressure distribution. Rarefaction effects (characterized by Kn) reduce the curvature in pressure distribution due to compressibility effects (characterized by Ma).

The first-order mass-flow relation (2.102) accurately predicts the experimental data available in the literature, for moderate outlet Knudsen numbers and typically for $Kn_o < 0.03$. For higher values of Kn_o , some deviations appear (cf. Section 2.3.5).

2.3.1.2. Second-order solutions

Therefore, the previous model can be improved to remain accurate for higher Knudsen numbers.

For this purpose, second-order – $\mathcal{O}(Kn^2)$ – models, based on second-order boundary conditions or the QHD equations, may be used.

The use of second-order boundary conditions

$$u_z|_{y=h} = -\frac{2-\sigma}{\sigma}\lambda \frac{\partial u_z}{\partial y}\bigg|_{y=h} + A_2\lambda^2 \frac{\partial^2 u_z}{\partial y^2}\bigg|_{y=h} \quad (2.107)$$

leads to a velocity distribution

$$u_z^* = 1 - y^{*2} + 8\frac{2-\sigma}{\sigma}Kn - 32A_2Kn^2 = 1 - y^{*2} + 4\frac{2-\sigma}{\sigma}Kn' - 8A_2Kn'^2 \quad (2.108)$$

and to a Poiseuille number

$$P_{\text{ONS2,plan}} = \frac{24}{1 + 12\frac{2-\sigma}{\sigma}Kn - 48A_2Kn^2} = \frac{24}{1 + 6\frac{2-\sigma}{\sigma}Kn' - 12A_2Kn'^2} \quad (2.109)$$

It is easy to verify that the mass flow rate is consequently corrected as:

$$\begin{aligned} \dot{m}_{\text{NS2,plan}}^* &= \frac{\dot{m}_{\text{NS2,plan}}}{\dot{m}_{\text{ns,plan}}} = 1 + 24\frac{2-\sigma}{\sigma}\frac{Kn_o}{\Pi + 1} - 96A_2Kn_o^2\frac{\ln \Pi}{\Pi^2 - 1} \\ &= 1 + 12\frac{2-\sigma}{\sigma}\frac{Kn'_o}{\Pi + 1} - 24A_2Kn'^2_o\frac{\ln \Pi}{\Pi^2 - 1} \end{aligned} \quad (2.110)$$

It is interesting to note that a close result can also be found keeping the first-order boundary conditions, but using the **QHD** equations in place of the **NS** equations (Elizarova and Sheretov, 2003). The solution of the momentum equation (2.40), with the expression (2.53) of the viscous stress tensor and the boundary condition (2.87) yields

$$\begin{aligned}\dot{m}_{\text{QHD1,plan}}^* &= \frac{\dot{m}_{\text{QHD1,plan}}}{\dot{m}_{\text{ns,plan}}} = 1 + 24 \frac{2-\sigma}{\sigma} \frac{\text{Kn}_0}{\Pi+1} + \frac{96}{k_2^2 \text{Sc}} \text{Kn}_0^2 \frac{\ln \Pi}{\Pi^2 - 1} \\ &= 1 + 12 \frac{2-\sigma}{\sigma} \frac{\text{Kn}'_0}{\Pi+1} + \frac{24}{k_2^2 \text{Sc}} \text{Kn}'_0{}^2 \frac{\ln \Pi}{\Pi^2 - 1}\end{aligned}\quad (2.111)$$

Equations (2.110) and (2.111) only differ with their last term.

For example, if we consider the second-order boundary conditions of Deissler, $A_2 = -9/8$ and $-96 A_2 = 108$. This value must be compared with $96/(k_2^2 \text{Sc})$. For an HS model, $k_2 = 16/5\sqrt{2\pi}$ and $\text{Sc} = 5/6$. Therefore, $96/k_2^2 \text{Sc} = 70, 7$. The deviation on the second-order term between the two models is then around 42%. But, if we consider the more accurate VSS model, with $\omega = 0.74$ for nitrogen under standard conditions (cf. Table 2.2), $k_2 = 1.064$, $\text{Sc} = 0.746$ and $96/k_2^2 \text{Sc} = 114$, which corresponds to a deviation of about 5% between the second-order terms of NS2 and QDH1 models. Of course, this deviation is a function of the gas, the temperature, the choice of the second-order boundary condition and of the IPL model.

2.3.2. Gas flow in circular microtubes

With the general assumptions of this section, the momentum equation (2.33) in polar coordinates reduces to

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{du_z}{dr} \right) = \frac{1}{\mu} \frac{dp}{dz} \quad (2.112)$$

2.3.2.1. First-order solution

The first-order boundary condition (2.58) for a non-moving wall and an isothermal flow is now:

$$u_z|_{r=r_2} = -\frac{2-\sigma}{\sigma} \lambda \left. \frac{\partial u_z}{\partial r} \right|_{r=r_2} \quad r_2 \quad (2.113)$$

(The inner and the outer radii are r_1 and r_2 , respectively. LK)

The general solution of Eq. (2.112) is

$$u_z = \frac{1}{\mu} \frac{dp}{dz} \left(\frac{r^2}{4} + c_1 \ln r + c_2 \right) \quad (2.114)$$

where $c_1 = 0$ since the velocity has a finite value in $r = 0$. Using the boundary condition (2.113), we find

$$c_2 = -\frac{2-\sigma}{\sigma} \lambda \frac{r_2}{2} - \frac{r_2^2}{4} \quad (2.115)$$

which leads to the dimensionless velocity profile

$$u_z^* = 1 - r^{*2} + 4 \frac{2-\sigma}{\sigma} \text{Kn} \quad (2.116)$$

where the Knudsen number $\text{Kn} = \lambda/D_H = \lambda/(2r_2) = \text{Kn}' = \lambda/L_{\min}$ is defined from a characteristic length $L = 2r_2$. Here, $r^* = r/r_2$ and $u_z^* = u_z/u_{z0}$ where

$$u_{z0} = \frac{r_2^2}{4\mu} \left(-\frac{dp}{dz} \right) \quad (2.117)$$

is the **velocity at the center** of the microchannel when rarefaction is not taken into account.

Equation (2.116) shows that the

usual **Hagen–Poiseuille velocity** profile ($u_z^* = 1 - r^{*2}$)
is just globally **translated** with a quantity

$2(2-\sigma)\lambda/(or_2)$ (or in dimensionless form $\frac{2-\sigma}{4}\text{Kn}$ LK)
due to the **slip at the wall**.

The integration of the velocity distribution given by Eq. (2.116) over the cross-section leads to the mean velocity

$$\overline{u_z^*} = \frac{1}{\pi} \int_0^1 u_z^* 2\pi r^* dr^* \quad (2.118)$$

that is with Eq. (2.116):

$$\overline{u_z^*} = \frac{1}{2} + 4 \frac{2-\sigma}{\sigma} \text{Kn} \quad (2.119)$$

The Poiseuille number Po is still defined by Eq. (2.99), which leads with Eqs. (2.117) and (2.119) to:

$$Po_{NS1,circ} = \frac{16}{1 + 8 \frac{2-\sigma}{\sigma} \text{Kn}} \quad (2.120)$$

If **Kn is not negligible**, the **Poiseuille number** is **less than** its usual value of **16** for a circular cross-section (Shah, 1975).

Following the same method as for a plane flow, the integration of Eq. (2.118) with $\dot{m} = p\overline{u_z}A/RT$ yields

$$\dot{m}_{\text{NS1,circ}} = \frac{\pi r_2^4 p_0^2}{4\mu RTl} \left[\frac{\Pi^2 - 1}{4} + 4 \frac{2 - \sigma}{\sigma} \text{Kn}_0 (\Pi - 1) \right] \quad (2.121)$$

for a long microtube, neglecting entrance effects and

$$\dot{m}_{\text{NS1,circ}}^* = \frac{\dot{m}_{\text{NS1,circ}}}{\dot{m}_{\text{ns,circ}}} = 1 + 16 \frac{2 - \sigma}{\sigma} \frac{\text{Kn}_0}{\Pi + 1} \quad (2.122)$$

The pressure distribution $p^*=p/p_0$ along the microtube is the solution of the polynomial

$$p^{*2} + \beta_1 p^* + \beta_2 + \beta_3 z^* = 0 \quad (2.123)$$

where $z^*=z/l$ and

$$\beta_1 = 16 \frac{2 - \sigma}{\sigma} \text{Kn}_0 \quad \beta_3 = (\Pi - 1) (\Pi + 1 + \beta_1) \quad \beta_2 = -1 - \beta_1 - \beta_3 \quad (2.124)$$

which qualitatively leads to the same conclusions as for a plane flow.

2.3.2.2. Second-order solution

The first-order solution can easily be extended to second-order boundary conditions, following the same reasoning as in Section 2.3.1.2.

$$u_z|_{y=h} = -\frac{2-\sigma}{\sigma}\lambda \frac{\partial u_z}{\partial y}\bigg|_{y=h} + A_2\lambda^2 \frac{\partial^2 u_z}{\partial y^2}\bigg|_{y=h} \quad (2.107)$$

We find

$$\overline{u_z^*} = \frac{1}{2} + 4\frac{2-\sigma}{\sigma}\text{Kn} - 8A_2\text{Kn}^2 \quad (2.125)$$

$$\text{PoNS2,circ} = \frac{16}{1 + 8\frac{2-\sigma}{\sigma}\text{Kn} - 16A_2\text{Kn}^2} \quad (2.126)$$

And

$$\dot{m}_{\text{NS2,circ}}^* = \frac{\dot{m}_{\text{NS2,circ}}}{\dot{m}_{\text{ns,circ}}} = 1 + 16\frac{2-\sigma}{\sigma} \frac{\text{Kn}_0}{\Pi + 1} - 32A_2\text{Kn}_0^2 \frac{\ln \Pi}{\Pi^2 - 1} \quad (2.127)$$

The problem can also be solved from the **QHD** equations with **first-order** boundary conditions. Lengrand *et al.* (2004) showed that

$$\dot{m}_{\text{QHD1,circ}}^* = \frac{\dot{m}_{\text{QHD1,circ}}}{\dot{m}_{\text{ns,circ}}} = 1 + 16\frac{2-\sigma}{\sigma} \frac{\text{Kn}_0}{\Pi + 1} + \frac{32}{k_2^2 \text{Sc}} \text{Kn}_0^2 \frac{\ln \Pi}{\Pi^2 - 1} \quad (2.128)$$

The **deviation** between the **QHD1** and the **NS2** models with respect to the second-order terms are **exactly the same as in the case of a plane flow** (Section 2.3.1.2).

2.3.3. Gas flow in annular ducts

The **solution** for the slip-flow of a gas in a

circular microduct

is easily extensible to the case of an

annular duct (Ebert and Sparrow, 1965).

The inner radius r_1 and the outer radius r_2 **are not necessarily small**, but the distance **$r_2 - r_1 = D_H/2$ must be small enough** so that the flow between the two cylinders is rarefied.

The first-order boundary conditions are **(Be careful to the + and – signs. LK)**

$$u_z = \frac{2-\sigma}{\sigma} \lambda \left. \frac{\partial u_z}{\partial r} \right|_{r=r_1} \quad u_z = -\frac{2-\sigma}{\sigma} \lambda \left. \frac{\partial u_z}{\partial r} \right|_{r=r_2} \quad (2.129)$$

The integration of Eq. (2.112) leads to the velocity distribution

$$\begin{aligned} \frac{1}{r} \frac{d}{dr} \left(r \frac{du_z}{dr} \right) &= \frac{1}{\mu} \frac{dp}{dz} \\ u_z^* &= 1 - r_1^{*2} + 4(1 - r_1^*) \frac{2-\sigma}{\sigma} \text{Kn} \\ &+ \frac{r_1^*(1 - r_1^{*2}) (1 + 4 \frac{2-\sigma}{\sigma} \text{Kn}) [2(1 - r_1^*) \frac{2-\sigma}{\sigma} \text{Kn} - \ln r_1^*]}{r_1^* \ln r_1^* - 2(1 - r_1^{*2}) \frac{2-\sigma}{\sigma} \text{Kn}} \end{aligned} \quad (2.130)$$

with $\text{Kn} = \text{Kn}'/2 = \lambda/(2(r_2 - r_1))$ and to its mean value

$$\begin{aligned} \overline{u_z^*} &= \frac{1}{2} \left[1 + r_1^{*2} + 8(1 - r_1^* + r_1^{*2}) \frac{2-\sigma}{\sigma} \text{Kn} \right. \\ &\quad \left. + \frac{r_1^*(1 - r_1^{*2})(1 + 4 \frac{2-\sigma}{\sigma} \text{Kn})^2}{r_1^* \ln r_1^* - 2(1 - r_1^{*2}) \frac{2-\sigma}{\sigma} \text{Kn}} \right] \end{aligned} \quad (2.131)$$

Thus, the Poiseuille number takes the form

$$\text{Po}_{\text{NS1, annu}} = \frac{16(1 - r_1^*)^2}{1 + r_1^{*2} + 8(1 - r_1^* + r_1^{*2}) \frac{2-\sigma}{\sigma} \text{Kn} + \frac{r_1^*(1 - r_1^{*2})(1 + 4 \frac{2-\sigma}{\sigma} \text{Kn})^2}{r_1^* \ln r_1^* - 2(1 - r_1^{*2}) \frac{2-\sigma}{\sigma} \text{Kn}}} \quad (2.132)$$

which generalizes the expression (2.120) obtained for a circular microtube.

2.3.4. Gas Flow in rectangular microchannels

With the rectangular cross-section defined in Fig. 2.10(c) and the general assumptions of Section 2.3, the momentum equation takes the form

$$\frac{d^2 u_z}{dx^2} + \frac{d^2 u_z}{dy^2} = \frac{1}{\mu} \frac{dp}{dz} \quad (2.133)$$

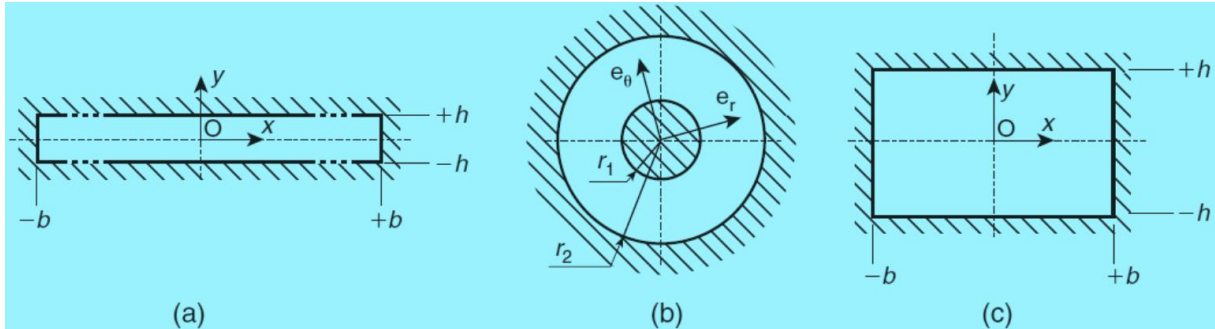
In a dimensionless form, the momentum equation can be written

$$a^{*2} \frac{d^2 u_z^*}{dx^{*2}} + \frac{d^2 u_z^*}{dy^{*2}} = -1 \quad (2.134)$$

with

$$x^* = \frac{x}{b} \quad y^* = \frac{y}{h} \quad u_z^* = \frac{u_z}{-\frac{h^2}{\mu} \frac{dp}{dz}} \quad (2.135)$$

and $a^* = h/b$ is the **aspect ratio** of the section with $0 < a^* \leq 1$, if we assume $h \leq b$.



2.3.4.1. First-order solution

The first-order boundary conditions are

$$u_z^*|_{y^*=1} = -\frac{2-\sigma}{\sigma} 2Kn' \left. \frac{\partial u_z^*}{\partial y^*} \right|_{y^*=1} \quad (2.136)$$

$$u_z^*|_{x^*=1} = -\frac{2-\sigma}{\sigma} 2a^*Kn' \left. \frac{\partial u_z^*}{\partial x^*} \right|_{x^*=1} \quad (2.137)$$

with $Kn' = \lambda/(2h)$ and the conditions of symmetry

$$\left. \frac{du_z^*}{dy^*} \right|_{y^*=0} = 0 \quad (2.138)$$

$$\left. \frac{du_z^*}{dx^*} \right|_{x^*=0} = 0 \quad (2.139)$$

Ebert and Sparrow (1965) proposed a solution for the velocity distribution, compatible with the condition of symmetry (2.138):

$$u_z^* = - \sum_{i=1}^{\infty} \Psi_i(x^*) \cos(\alpha_i y^*) \quad (2.140)$$

The eigenvalues α_i must verify

$$\alpha_i \tan \alpha_i = \frac{\sigma}{2-\sigma} \frac{1}{2Kn'} \quad (2.141)$$

in order to be compatible with the boundary condition (2.136) without leading to a trivial velocity solution.

The x^* dependent ψ_i functions are found by expanding the right-hand side of Eq. (2.134), that is unity, in terms of the orthogonal functions $\cos(\alpha_i y^*)$:

$$a^{*2} \frac{d^2 u_z^*}{dx^{*2}} + \frac{d^2 u_z^*}{dy^{*2}} = -1 \quad (2.134)$$

$$-1 = - \sum_{i=1}^{\infty} \Omega_i \cos(\alpha_i y^*) = - \sum_{i=1}^{\infty} \frac{2 \sin \alpha_i}{\alpha_i + \sin \alpha_i \cos \alpha_i} \cos(\alpha_i y^*) \quad (2.142)$$

Therefore, to verify the momentum equation (2.134), the functions ψ_i must obey

$$\frac{d^2 \Psi_i}{dx^{*2}} - \left(\frac{\alpha_i}{a^*} \right)^2 \Psi_i - \frac{\Omega_i}{a^{*2}} = 0 \quad (2.143)$$

The solution of Eq. (2.143), with the conditions (2.137)–(2.139) and using Eq. (2.141), is

$$u_z^*|_{x^*=1} = -\frac{2-\sigma}{\sigma} 2a^* \text{Kn}' \left. \frac{\partial u_z^*}{\partial x^*} \right|_{x^*=1} \quad (2.137)$$

$$\left. \frac{du_z^*}{dy^*} \right|_{y^*=0} = 0 \quad (2.138)$$

$$\left. \frac{du_z^*}{dx^*} \right|_{x^*=0} = 0 \quad (2.139)$$

$$\Psi_i = -\frac{\Omega_i}{\alpha_i^2} \left(1 - \frac{\cosh\left(\frac{\alpha_i x^*}{a^*}\right)}{\cosh\left(\frac{\alpha_i}{a^*}\right) + 2\text{Kn}' \frac{2-\sigma}{\sigma} \alpha_i \sinh\left(\frac{\alpha_i}{a^*}\right)} \right) \quad (2.144)$$

Consequently, the velocity distribution is given by Eq. (2.140).

This solution is given as a function of a Knudsen number $\text{Kn}' = \lambda/(2h)$ based on the channel depth, but it can also be expressed as a function of the previous Knudsen number $\text{Kn} = \lambda/D_H$ based on the hydraulic diameter by using the relation

$$2\text{Kn} = \text{Kn}'(1 + a^*) \quad (2.145)$$

In the special case of a square section, for which $a^*=1$, both Knudsen numbers Kn and Kn' are identical.

2.3.4.2. Second-order solution

The above solution from Ebert et Sparrow was improved by Aubert and Colin (2001), in order to be more accurate for higher Knudsen numbers. This model is based on the second-order boundary conditions proposed by Deissler (1964):

$$u_z|_{y=h} = -\frac{2-\sigma}{\sigma}\lambda \left. \frac{\partial u_z}{\partial y} \right|_{y=h} - \frac{9}{16}\lambda^2 \left(2 \left. \frac{\partial^2 u_z}{\partial y^2} \right|_{y=h} + \left. \frac{\partial^2 u_z}{\partial x^2} \right|_{y=h} + \left. \frac{\partial^2 u_z}{\partial z^2} \right|_{y=h} \right) \quad (2.146)$$

and

$$u_z|_{x=b} = -\frac{2-\sigma}{\sigma}\lambda \left. \frac{\partial u_z}{\partial x} \right|_{x=b} - \frac{9}{16}\lambda^2 \left(2 \left. \frac{\partial^2 u_z}{\partial x^2} \right|_{x=b} + \left. \frac{\partial^2 u_z}{\partial y^2} \right|_{x=b} + \left. \frac{\partial^2 u_z}{\partial z^2} \right|_{x=b} \right) \quad (2.147)$$

This choice of Deissler boundary conditions, rather than other second-order formulations such as the ones proposed in Section 2.2.3.3, is due to the physical approach proposed by Deissler. The above equations are obtained from a three dimensional local momentum balance at the wall and the shear stress satisfies the definition of a Newtonian fluid used in the Navier–Stokes equations.

In Eqs. (2.146) and (2.147), the term $\partial^2 u_z / \partial z^2$ vanishes according to the continuity equation, which reduces to $\partial u_z / \partial z = 0$ for a locally fully developed flow.

Finally, using the momentum equation (2.134), these equations may be written in a non-dimensional form as follows:

$$a^{*2} \frac{d^2 u_z^*}{dx^{*2}} + \frac{d^2 u_z^*}{dy^{*2}} = -1 \quad (2.134)$$

$$u_z^*|_{y^*=1} = -\frac{2-\sigma}{\sigma} 2Kn' \left. \frac{\partial u_z^*}{\partial y^*} \right|_{y^*=1} - \frac{9}{4}Kn'^2 \left(\left. \frac{\partial^2 u_z^*}{\partial y^{*2}} \right|_{y^*=1} - 1 \right) \quad (2.148)$$

and

$$u_z^*|_{x^*=1} = -\frac{2-\sigma}{\sigma} 2a^*Kn' \left. \frac{\partial u_z^*}{\partial x^*} \right|_{x^*=1} - \frac{9}{4}Kn'^2 \left(a^{*2} \left. \frac{\partial^2 u_z^*}{\partial x^{*2}} \right|_{x^*=1} - 1 \right) \quad (2.149)$$

The velocity distribution (2.140) proposed by Ebert and Sparrow,

$$u_z^* = - \sum_{l=1}^{\infty} \Psi_l(x^*) \cos(\alpha_l y^*) \quad (2.140)$$

in the form of a single Fourier series, **does not converge** with **second-order boundary conditions**. Actually, the eigenvalues α_i obtained from the boundary condition (2.148) are such that the functions $\cos(\alpha_i y^*)$ are no longer orthogonal on the interval $[-1, 1]$. Therefore, a **new form**

$$u_z^*(x^*, y^*) = \sum_{i,j=1}^{\infty} A_{ij} N_{ij} \cos\left(\varphi_i \frac{x^*}{a^*}\right) \cos(\psi_j y^*) + \frac{9}{4} \text{Kn}'^2 \quad (2.150)$$

is required, based on a double Fourier series.

The functions $\omega_{ij}(x^*, y^*) = \cos(\varphi_i x^*/a^*) \cos(\psi_j y^*)$ are orthogonal over the domain $[-1, 1] \times [-1, 1]$. They are solutions to the eigenvalues problem stemming from Eqs. (2.134), (2.148) and (2.149).

$$a^{*2} \frac{d^2 u_z^*}{dx^{*2}} + \frac{d^2 u_z^*}{dy^{*2}} = -1 \quad (2.134)$$

$$u_z^*|_{y^*=1} = -\frac{2-\sigma}{\sigma} 2\text{Kn}' \frac{\partial u_z^*}{\partial y^*} \Big|_{y^*=1} - \frac{9}{4} \text{Kn}'^2 \left(\frac{\partial^2 u_z^*}{\partial y^{*2}} \Big|_{y^*=1} - 1 \right) \quad (2.148)$$

$$u_z^*|_{x^*=1} = -\frac{2-\sigma}{\sigma} 2a^* \text{Kn}' \frac{\partial u_z^*}{\partial x^*} \Big|_{x^*=1} - \frac{9}{4} \text{Kn}'^2 \left(a^{*2} \frac{\partial^2 u_z^*}{\partial x^{*2}} \Big|_{x^*=1} - 1 \right) \quad (2.149)$$

The terms N_{ij} are such that the functions $N_{ij} \cos(\varphi_i x^*/a^*) \cos(\psi_j y^*)$ are normed over the domain $[-1, 1] \times [-1, 1]$.

The conditions (2.148) and (2.149) require ψ_j and φ_i to be solutions of Aubert (1999):

$$1 - 2\text{Kn}' \frac{(2-\sigma)}{\sigma} \psi_j \tan \psi_j - \frac{9}{4} \text{Kn}'^2 \psi_j^2 = 0 \quad (2.151)$$

and

$$1 - 2\text{Kn}' \frac{(2-\sigma)}{\sigma} \varphi_i \tan \frac{\varphi_i}{a^*} - \frac{9}{4} \text{Kn}'^2 \varphi_i^2 = 0 \quad (2.152)$$

In order to determine the terms A_{ij} , the solution (2.150) is injected into the momentum equation (2.134). Furthermore, the right-hand side of Eq. (2.134) is expanded in a double Fourier series. Finally, it is found that

$$\begin{aligned}
A_{ij}N_{ij} = & \frac{1}{\varphi_i^2 + \psi_j^2} \left\{ \frac{4 \sin(\varphi_i/a^*) \sin \psi_j}{\varphi_i \psi_j} + \frac{9}{2} \frac{\sigma}{2 - \sigma} \text{Kn}' \left(\frac{\sin(\varphi_i/a^*) \cos \psi_j}{\varphi_i} + \frac{\cos(\varphi_i/a^*) \sin \psi_j}{\psi_j} \right) \right\} \\
& \times \left\{ \left(\frac{1}{a^*} + \frac{\cos(\varphi_i/a^*) \sin(\varphi_i/a^*)}{\varphi_i} \right) \left(1 + \frac{\sin \psi_j \cos \psi_j}{\psi_j} \right) \right. \\
& + \frac{9}{4} \frac{\sigma}{2 - \sigma} \text{Kn}' \left(\cos^2 \psi_j \left[\frac{1}{a^*} + \frac{\cos(\varphi_i/a^*) \sin(\varphi_i/a^*)}{\varphi_i} \right] \right. \\
& \left. \left. + \cos^2(\varphi_i/a^*) \left[1 + \frac{\sin \psi_j \cos \psi_j}{\psi_j} \right] \right) \right\}^{-1} \quad (2.153)
\end{aligned}$$

Whatever the aspect ratio a^* , the convergence of the series (2.150) was verified. In every case, the asymptotic value of the sum is reached with a good precision from a number of roots ψ_j (Kn' , Kn'^2) and φ_i (a^* , Kn' , Kn'^2) from Eqs. (2.151) and (2.152) lower than 50.

In **Figure 2.12**, the normalized velocity distribution u_z^* is plotted on the axis $x^*=0$.

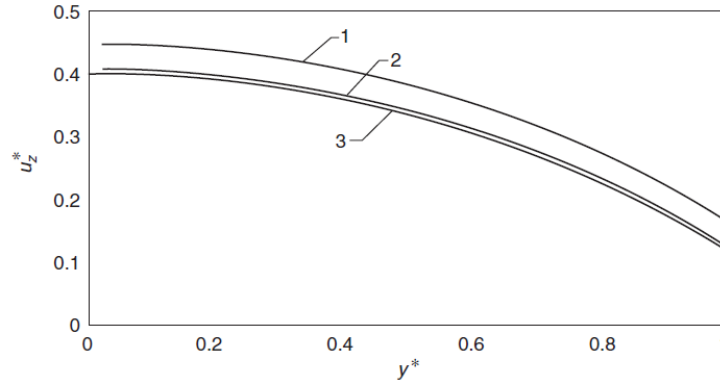


Fig. 2.12. Normalized velocity distribution in a square cross-section with diffuse reflection at the wall for $\text{Kn}'=0.1$.

- 1: second-order model;
- 2: first-order model (Ebert and Sparrow, 1965);
- 3: first-order model (Morini and Spiga, 1998).

The solutions proposed by Morini and Spiga (1998) and by Ebert and Sparrow (1965) and the second-order solution of Eq. (2.150) are compared for $\sigma = 1$, $\text{Kn}'=0.1$ and for a square section ($a^* = 1$). The slight deviation observed between the velocity distributions found by Morini and Spiga and by Ebert and Sparrow results from their two different ways of solving the same first-order problem. It is pointed out that the flow rate is underestimated (about 10% in the case of **Fig. 2.12**) when the second-order terms are not taken into account. Moreover, it is

verified that the velocity distribution obtained with a second-order model cannot be deduced from a simple translation of the first-order velocity distribution.

The mass flow rate through the cross-section is

$$\dot{m} = \iint_{(\Sigma)} \rho u_z dx dy$$

that is with Eqs. (2.135) and (2.150):

$$x^* = \frac{x}{b} \quad y^* = \frac{y}{h} \quad u_z^* = \frac{u_z}{-\frac{h^2}{\mu} \frac{dp}{dz}} \quad (2.135)$$

$$u_z^*(x^*, y^*) = \sum_{i,j=1}^{\infty} A_{ij} N_{ij} \cos\left(\psi_i \frac{x^*}{a^*}\right) \cos(\psi_j y^*) + \frac{9}{4} \text{Kn}'^2 \quad (2.150)$$

$$\dot{m}_{\text{NS2,rect}} = -\frac{\rho h^2}{\mu} \frac{dp}{dz} \frac{4h^2}{a^*} \left[\sum_{i,j=1}^{\infty} A_{ij} N_{ij} \frac{a^* \sin(\varphi_i/a^*) \sin \psi_j}{\varphi_i \psi_j} + \frac{9}{4} \text{Kn}'^2 \right] \quad (2.154)$$

In order to obtain $\dot{m}$ as a function of the inlet pressure p_i and the outlet pressure p_o of a long rectangular microchannel of length l , Eq. (2.154) must be integrated along the microchannel. For an isothermal flow and known geometric dimensions, the Knudsen number only depends on the pressure $p(z)$. Therefore, the solutions ϕ_i and ψ_j of Eqs. (2.151) and (2.152) depend on the position along the z -axis. The bracketed term of Eq. (2.154) can be precisely fitted by a polynomial function $a_1 + a_2 \text{Kn}' + a_3 \text{Kn}'^2$, where a_1 , a_2 and a_3 are coefficients depending on the aspect ratio a^* of the cross-section and on the momentum accommodation coefficient σ for a_2 and a_3 . Thus,

$$\dot{m}_{\text{NS2,rect}} = -\frac{4ph^4}{a^* RT \mu} \frac{dp}{dz} [a_1 + a_2 \text{Kn}' + a_3 \text{Kn}'^2] \quad (2.155)$$

Some numerical values of the coefficients a_1 , a_2 and a_3 , obtained with a least squares fitting method from values calculated using Eq. (2.154) for $\text{Kn}' \in [0; 1]$, are given in Table 2.5. Note that the numerical values of the coefficient a_1 correspond closely to the analytical expression

$$a_1 = 2 \sum_{i=1}^{\infty} \frac{\sin^2 \alpha_{i,\text{ns}}}{\alpha_{i,\text{ns}}^4 (\alpha_{i,\text{ns}} + \cos \alpha_{i,\text{ns}} \sin \alpha_{i,\text{ns}})} \left(\alpha_{i,\text{ns}} - a^* \tanh \frac{\alpha_{i,\text{ns}}}{a^*} \right) \quad (2.156)$$

which can be deduced from the no-slip problem. In Eq. (2.156), $\alpha_{i,\text{ns}} = (2i-1)\pi/2$.

Table 2.5

Values of coefficients a_1 , a_2 and a_3 for different values of σ and a^* , for Deissler second-order boundary conditions.

a^*	σ	0.5	0.6	0.7	0.8	0.9	1
0		$a_2 = 6.0000$	$a_2 = 4.6667$	$a_2 = 3.7143$	$a_2 = 3.0000$	$a_2 = 2.4444$	$a_2 = 2.0000$
$a_1 = 0.33333$		$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$
0.01		$a_2 = 5.9694$	$a_2 = 4.6459$	$a_2 = 3.7007$	$a_2 = 2.9917$	$a_2 = 2.4405$	$a_2 = 1.9993$
$a_1 = 0.33123$		$a_3 = 4.4528$	$a_3 = 4.4542$	$a_3 = 4.4548$	$a_3 = 4.4550$	$a_3 = 4.4549$	$a_3 = 4.4549$
0.025		$a_2 = 5.9228$	$a_2 = 4.6141$	$a_2 = 3.6797$	$a_2 = 2.9788$	$a_2 = 2.4337$	$a_2 = 1.9977$
$a_1 = 0.32808$		$a_3 = 4.3840$	$a_3 = 4.3875$	$a_3 = 4.3887$	$a_3 = 4.3895$	$a_3 = 4.3896$	$a_3 = 4.3892$
0.05		$a_2 = 5.8420$	$a_2 = 4.5591$	$a_2 = 3.6423$	$a_2 = 2.9547$	$a_2 = 2.4201$	$a_2 = 1.9925$
$a_1 = 0.32283$		$a_3 = 4.2761$	$a_3 = 4.2813$	$a_3 = 4.2843$	$a_3 = 4.2858$	$a_3 = 4.2858$	$a_3 = 4.2853$
0.1		$a_2 = 5.6774$	$a_2 = 4.4431$	$a_2 = 3.5613$	$a_2 = 2.9000$	$a_2 = 2.3860$	$a_2 = 1.9751$
$a_1 = 0.31233$		$a_3 = 4.0697$	$a_3 = 4.0832$	$a_3 = 4.0906$	$a_3 = 4.0943$	$a_3 = 4.0952$	$a_3 = 4.0944$
0.25		$a_2 = 5.1332$	$a_2 = 4.0464$	$a_2 = 3.2675$	$a_2 = 2.6823$	$a_2 = 2.2267$	$a_2 = 1.8626$
$a_1 = 0.28081$		$a_3 = 3.6073$	$a_3 = 3.6287$	$a_3 = 3.6449$	$a_3 = 3.6560$	$a_3 = 3.6626$	$a_3 = 3.6647$
0.5		$a_2 = 4.2443$	$a_2 = 3.3519$	$a_2 = 2.7120$	$a_2 = 2.2289$	$a_2 = 1.8487$	$a_2 = 1.5395$
$a_1 = 0.22868$		$a_3 = 3.3489$	$a_3 = 3.3588$	$a_3 = 3.3708$	$a_3 = 3.3856$	$a_3 = 3.4048$	$a_3 = 3.4292$
1		$a_2 = 3.1207$	$a_2 = 2.4552$	$a_2 = 1.9782$	$a_2 = 1.6187$	$a_2 = 1.3364$	$a_2 = 1.1077$
$a_1 = 0.14058$		$a_3 = 3.3104$	$a_3 = 3.3176$	$a_3 = 3.3287$	$a_3 = 3.3438$	$a_3 = 3.3644$	$a_3 = 3.3908$

Since for an isothermal flow, the quantity $\text{Kn}'p$ is constant, the integration of Eq. (2.155) along the microchannel yields

$$\dot{m}_{\text{NS2,rect}} = \frac{4h^4 p_o^2}{a^* \mu R T l} \left[\frac{a_1}{2} (\Pi^2 - 1) + a_2 \text{Kn}'_o (\Pi - 1) + a_3 \text{Kn}_o'^2 \ln \Pi \right] \quad (2.157)$$

and consequently

$$\dot{m}_{\text{NS2,rect}}^* = \frac{\dot{m}_{\text{NS2,rect}}}{\dot{m}_{\text{ns,rect}}} = \left[1 + 2 \frac{a_2}{a_1} \text{Kn}'_o \frac{1}{\Pi + 1} + 2 \frac{a_3}{a_1} \text{Kn}_o'^2 \frac{\ln \Pi}{\Pi^2 - 1} \right] \quad (2.158)$$

The mass flow rate is strongly dependent on the aspect ratio (cf. Fig. 2.13). If a^* is less than 0.01, the simpler plane flow model of Section 2.3.1 is accurate enough, but for higher values of a^* , it is no longer the case and the model presented in this section should be used.

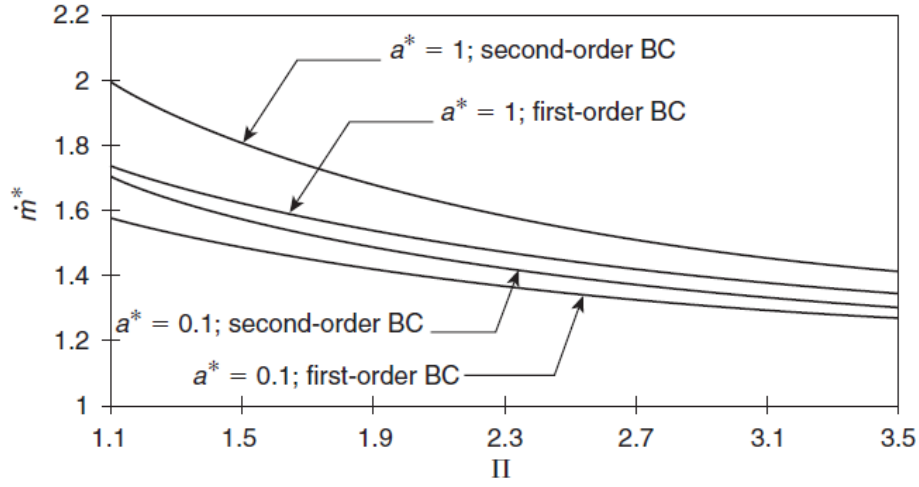


Fig. 2.13. Influence of the aspect ratio and of the slip-flow model on the normalized mass flow rate in a rectangular microchannel, for $Kn'_0=0.1$ and $\sigma=1$. BC: boundary condition.

As an example, let us consider a pressure-driven flow in a long rectangular microchannel with an inlet over outlet pressure ratio $\Pi=2$, an outlet Knudsen number $Kn'_0=0.1$ and an accommodation coefficient $\sigma=1$. The flow rate deduced from Eq. (2.110)

$$\dot{m}_{NS2,plan} = [(2bh^3p_0^2)/(\mu RTl)][(\Pi^2 - 1)/3 + 4((2 - \sigma)/\sigma)Kn'_0(\Pi - 1) + 9Kn_0'^2 \ln \Pi]$$

which was calculated from a plane model with Deissler boundary conditions, overestimates the flow rate $\dot{m}_{NS2,plan}$ given by Eq. (2.157) and calculated from a rectangular model. The overestimation $[(\dot{m}_{plan} - \dot{m}_{rect})/\dot{m}_{rect}]_{NS2}$ is about 0.5% for an aspect ratio $a^*=0.01$, it rises to 5.3% for an aspect ratio $a^*=0.1$ and reaches 112% for a square cross-section with $a^*=1$!

The **pressure distribution $p(z)$** can also be deduced from the **mass conservation**. It is the solution of the equation

$$\frac{\frac{a_1}{2}[(p/p_o)^2 - 1] + a_2 \text{Kn}'_o[(p/p_o) - 1] + a_3 \text{Kn}'_o{}^2 \ln(p/p_o)}{\left[\frac{a_1}{2}(\Pi^2 - 1) + a_2 \text{Kn}'_o(\Pi - 1) + a_3 \text{Kn}'_o{}^2 \ln \Pi\right]} = (1 - z^*) \quad (2.159)$$

which can be numerically solved. An example of the non-dimensional pressure distribution $p^*=p/p_{\text{ns}}$ as a function of $z^*=z/l$ is given in Fig. 2.14, for an outlet Knudsen number $\text{Kn}'_o=0.1$, an accommodation coefficient $\sigma=1$ and for $a^*=1$ (square duct) or $a^*=0.1$. The maximum influence of the slip at the wall corresponds to $z^*=0.77$. A no-slip model overestimates the pressure by a few percents, as is the case for a plane microchannel (Fig. 2.11). Moreover, the greater the aspect ratio, the larger the pressure overestimation is.

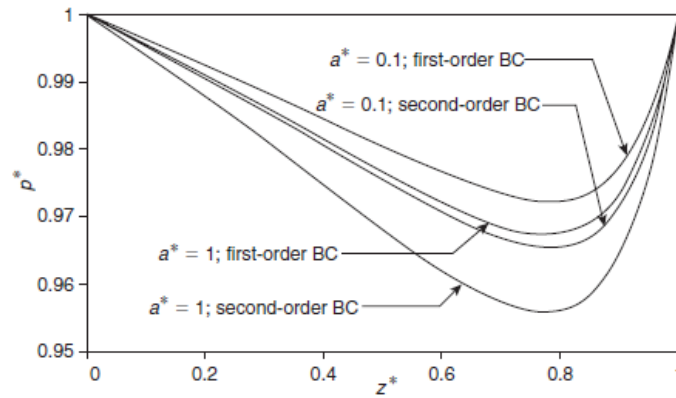


Fig. 2.14. Influence of the aspect ratio and of the slip-flow model on the normalized pressure distribution along a rectangular microchannel, for $\text{Kn}'_o=0.1$ and $\sigma=1$. BC: boundary condition.

2.3.5. Experimental data

A discussion of the accuracy of the different models described in the previous section requires fine experimental data, notably to know which second-order boundary conditions are more appropriate.

2.3.5.1. Experimental setups for flow rate measurements

Few experimental setups are described in the literature. The first techniques used to provide experimental data of gas flow rates in a microchannel were proposed by Arkilic *et al.* (1994) or by Pong *et al.* (1994). They both used accumulation techniques: the gas flowing through the microchannel accumulates in a reservoir, resulting in either a change in pressure (constant-volume technique) or a change in volume (constant-pressure technique). Actually, this method is limited because the tiny mass flows are very difficult to measure accurately using a single tank. In particular very small changes in temperature can overwhelm the mass flow measurement due to thermal expansion of the gas. To overcome this difficulty, a differential technique was used afterwards (Arkilic *et al.*, 1998). The system is schematically represented in Fig. 2.15. Gas flows through the test microchannel and into the accumulation tank A. The mass flow through the system can be inferred from a measurement of the differential pressure between the accumulation tank A and a reference tank B. This arrangement is very insensitive to temperature fluctuations and allows accurate measurements of mass flow rates, with a sensitivity announced to be as low as 7×10^{-5} kg/s.

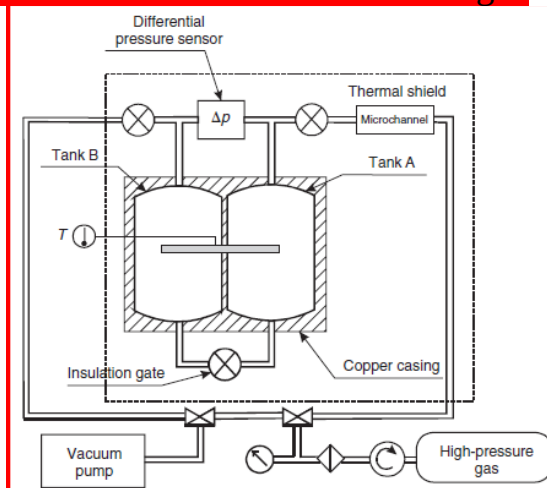


Fig. 2.15. Schematic of an accumulation mass flow measurement system. Adapted from Arkilic *et al.* (1998).

Another experimental setup was designed by Lalonde (2001) for the measurement of gaseous microflow rates under controlled temperature and pressure conditions, in the range 10^{-7} to 10^{-13} m³/s. Its principle is based on the tracking of a drop meniscus in a calibrated pipette connected in series with the microchannel. Its main advantage, compared to other similar setups presented in the literature (Shih

et al., 1996; Zohar *et al.*, 2002; Maurer *et al.*, 2003), is the simultaneous measurement of the flow rate both upstream and downstream of the microchannel (Fig. 2.16). The inlet and the outlet pressures can both be independently tuned. The volume flow rate is measured by means of opto-electronical sensors that detect the passage of the two menisci of a liquid drop injected into the calibrated pipettes connected upstream and downstream of the microchannel.

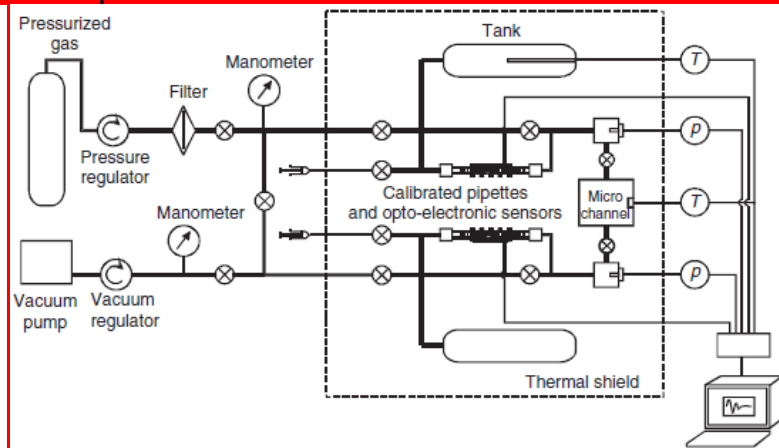


Fig. 2.16. Schematic of a pointer tracking system for the measurement of gas microflow rates (Lalonde, 2001).

A series of individual measurements of the volumetric flow rate can therefore be obtained, which allows the determination of the mean volumetric flow rates upstream and downstream of the microchannel and consequently, the mass flow rate through the microchannel. The comparison of the individual data given by each pipette, as well as the comparison of the two mean mass flow rates deduced from the two mean volumetric flow rates, must be consistent with the experimental uncertainties to allow the validation of the acquisition. A typical example is shown in Fig. 2.17. The upper data (circles) correspond to the downstream pipette and the lower ones (triangles) to the upstream pipette, through which the volumetric flow rate is lower, since the pressure is higher. The white symbols are relative to the first, the dark symbols to the second, of the two menisci of the drop in the pipette. In total, 44 individual data values are obtained for each test. Experimental uncertainties are reported by vertical bars.

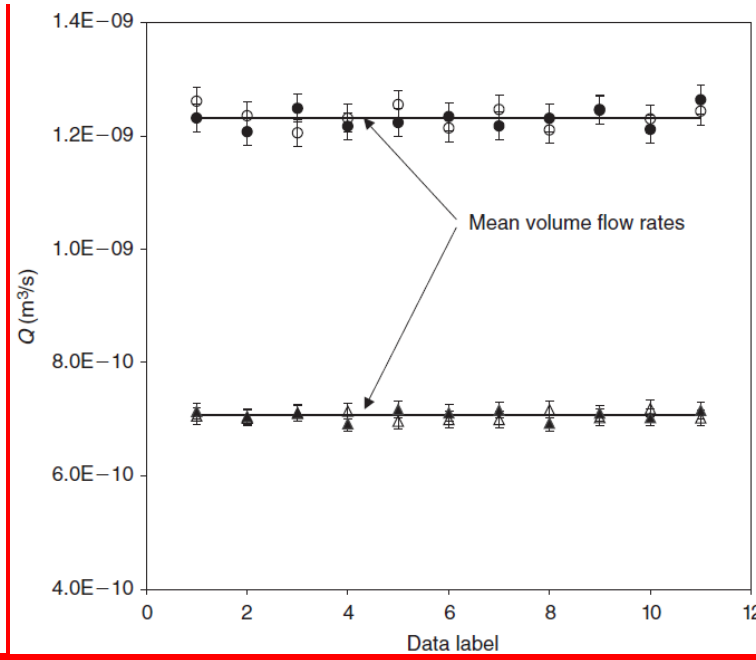


Fig. 2.17. Typical example of volumetric flow rate (Q) measurement with the setup of Lalonde (2001).

Other similar setups are described by Zohar *et al.* (2002) or by Maurer *et al.* (2003). They both provide a single but accurate measurement downstream of the microchannel. In the first case (Zohar *et al.*, 2002), the outlet pressure is always the atmospheric pressure, whereas in the second case (Maurer *et al.*, 2003), the outlet can be tuned as well as the upstream pressure.

2.3.5.2. Flow rate data

Comparing the different models available to describe the slip flow regime in microchannels, we note significant differences, essentially for the second-order term. For example, in a plane flow between parallel plates, the mass flow rate predicted by Eq. (2.110) depends on the choice of the second-order boundary condition. According to the value of the coefficient A_2 (see Table 2.3), the calculated mass flow rate may vary significantly. Moreover, some models generally based on a simple mathematical extension of Maxwell's condition (2.58) and for which $A_2 > 0$ predict a decrease of the slip compared to the first-order model. While other models, for which $A_2 < 0$, predict an increase of the slip. It is also interesting to test the accuracy of a QHD model, which does not involve second-order boundary conditions, but is second-order Knudsen number dependent (*cf.* Eq. (2.111)).

The increase of the slip due to second-order terms was first confirmed by Lalonde (2001), who measured the flow rate of helium and nitrogen in rectangular microchannels and showed that for outlet Knudsen numbers $Kn'_{0,M}$ higher than

0.05, a first-order model underestimates the slip. For the comparison of the experimental flow rate data with the predictions from the different models presented above, the sources of error or uncertainty must be considered. They can globally be divided into three classes:

(1) The uncertainties inherent to the fluid properties (density, viscosity, dissolved air, etc.) and to their dependence toward the experimental parameters (temperature, pressure). One can also note that some properties such as the momentum accommodation coefficient, σ , depend both on the fluid and the wall.

(2) The uncertainties relative to the microchannel geometrical characteristics (dimensions of the section, roughness, etc.) and to their dispersion.

(3) The uncertainties due to the flow rate measurement (metrology, leakage, operating conditions, etc.).

Most of these uncertainties can be limited within an acceptable precision. This is generally not the case for the measurement of the cross-section dimensions. When these dimensions are of the order of $1\mu\text{m}$, the measures obtained by different means (profilometer, optical microscope, SEM) that require preliminary calibration can differ notably: up to about 5% for the width and about 10% for the depth (Anduze, 2000). Therefore, the data are hardly exploitable, since the hydraulic diameter plays a part to the power of 4 in the calculation of the flow rate. Also, an inaccuracy of only 10 nm when measuring a depth of $1\mu\text{m}$ leads to an error of 4% in the estimation of the flow rate. The second parameter that can pose a problem is the accommodation coefficient σ , the value of which is not usually well known.

In order to avoid interpretation errors due to these two main sources of uncertainty, the idea is to compare among themselves data measured on a same microchannel, in a Knudsen number range that covers at least two of the following regimes:

(1) In the *little rarefied* regime, differences between the flow rates predicted by the no-slip (ns), the NS1 and the NS2 models are negligible, of the order of the experimental uncertainties. The value of σ consequently does not play any significant role.

(2) In the *lightly rarefied* regime, the difference between NS1 and NS2 models remains negligible, but the difference with the ns model is now significant. The influence of the accommodation coefficient σ may also become significant for the higher values of Kn'_0 .

(3) In the *moderately rarefied* regime, the ns, NS1 and NS2 models predict significantly different flow rates, and σ is now a very sensitive parameter.

Figure 2.18 represents these regimes with a series of microchannels experimentally studied by Lalonde (Colin *et al.*, 2004).

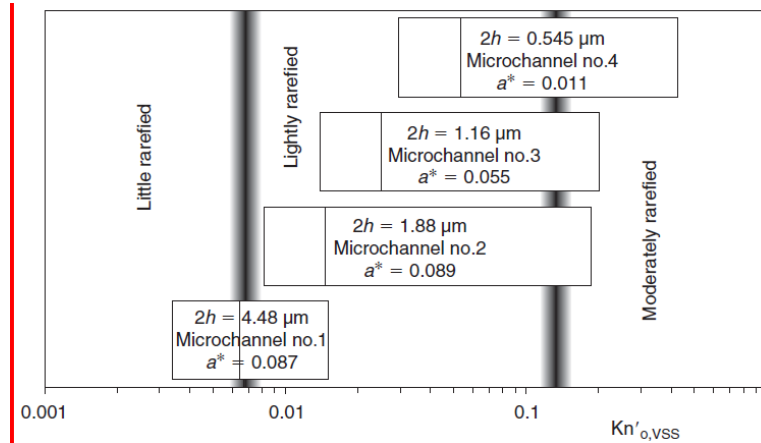


Fig. 2.18. Knudsen number ranges covered by a series of rectangular microchannels tested by Lalonde (2001), the right part of the rectangles corresponds to the outlet Knudsen number values.

The microchannel no.1, with a measured depth $2h=4.48\mu\text{m}$, allows a check of the transition from ns to NS1 models. First, the flow rate is measured for low values of the outlet Knudsen number Kn'_0 , that is for high values of P_0 . For $\text{Kn}'_{0,VSS}=0.007$, the differences between the flow rates predicted from the NS1 and NS2 models are negligible, whatever the value of the accommodation coefficient σ and the deviation with the ns model remains of the order of the experimental uncertainties. The depth $2h$ kept for the simulation is adjusted to $4.48\mu\text{m}$ – exactly the measured value in that case – for a good correlation between the experimental data and the NS1 or NS2 model. Then, rarefaction is increased, by decreasing the outlet pressure; the difference between ns and NS1 or NS2 models becomes significant and the experimental data are in very good agreement with the slip flow models. The influence of σ is still negligible as is the deviation between NS1 and NS2 models.

The microchannels nos.3 and 4 allow a check of the transition from NS1 to NS2 models. As an illustration, Fig. 2.19 shows some flow rate data relative to the microchannel no.3, whose measured depth is $2h=1.15\mu\text{m}$. The depth for the simulation is adjusted to $1.16\mu\text{m}$, that is $0.01\mu\text{m}$ more than the measured value, from data at low values of Kn'_0 , for which both NS1 and NS2 models accurately predict the experimental flow rates. An increase of the rarefaction leads to a deviation between these two models. The agreement between the experimental data and the NS2 model is good, whatever the gas – nitrogen (Fig. 2.19(b)) or helium (Fig. 2.19(c)) – with $\sigma=0.93$. The rarefaction is increased even more and

the difference between NS1 and NS2 models becomes very significant (Fig. 2.19(d)). The experiment shows the validity of the second-order slip flow model with Knudsen numbers higher than 0.2. This observation is confirmed with the microchannel no.4, up to Knudsen numbers around 0.25.

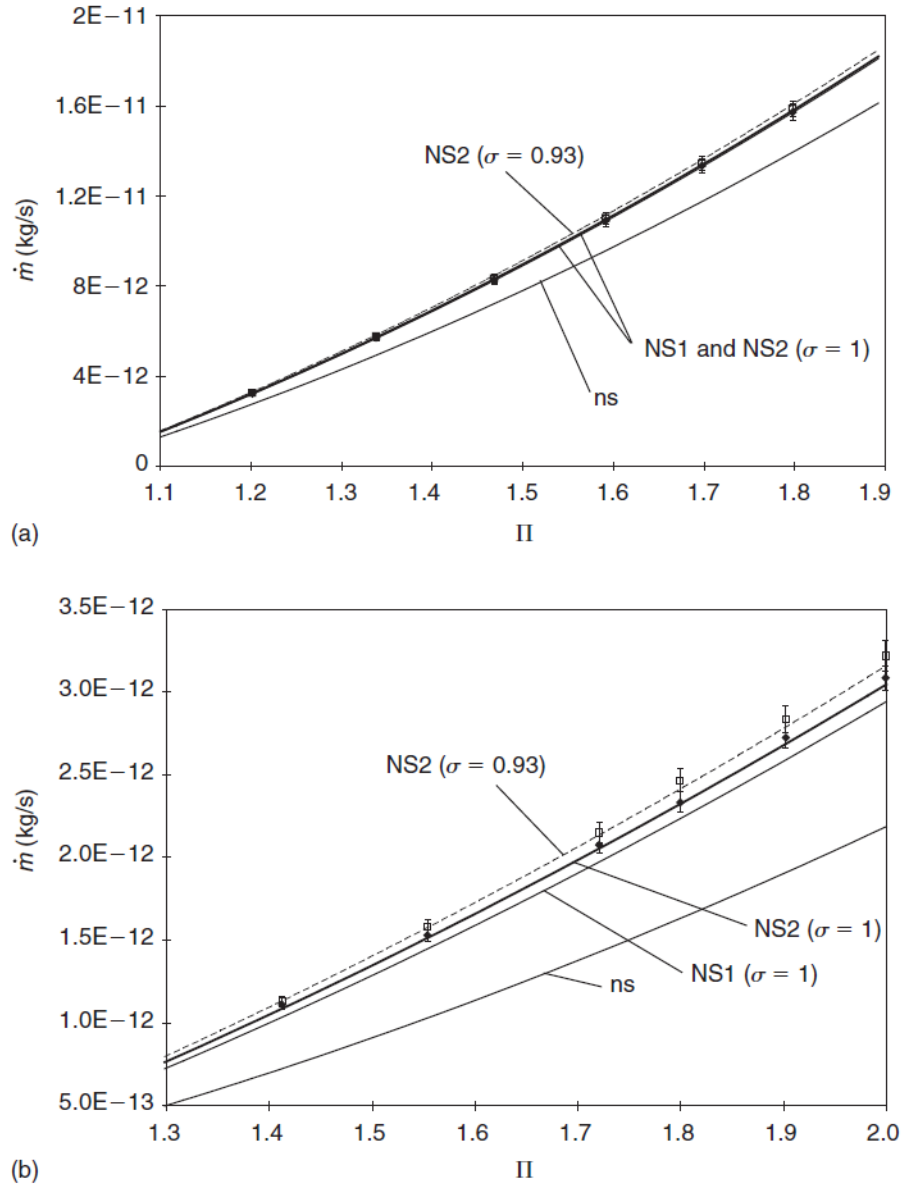


Fig. 2.19. Theoretical and experimental flow rates for a rectangular microchannel. 2 $h=1.16\mu\text{m}$, $2b=21\mu\text{m}$, $l=5000\mu\text{m}$, $T=294.2$ K. (a) Gas: N₂, $p_0 = 1.9 \times 10^5$ Pa, $\text{Kn}'_{0,VSS} = 2.5 \times 10^{-2}$; (b) gas: N₂, $p_0 = 0.65 \times 10^5$ Pa, $\text{Kn}'_{0,VSS} = 7.3 \times 10^{-2}$.

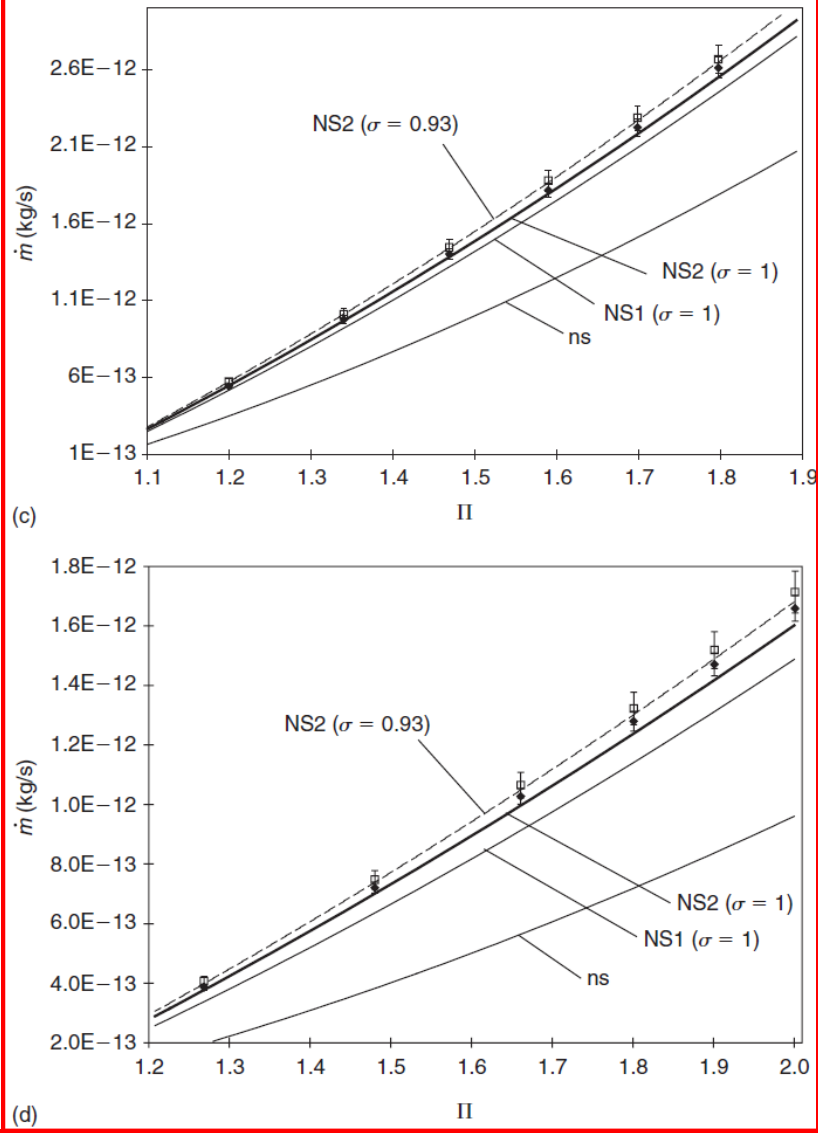


Fig. 2.19. (Continued) (c) gas: He, $p_o = 1.9 \times 10^5$ Pa, $Kn'_{0,VSS} = 7.9 \times 10^{-2}$; (d) gas: He, $p_o = 0.75 \times 10^5$ Pa, $Kn'_{0,VSS} = 2.0 \times 10^{-1}$.

For nitrogen and helium flows, an attempt to summarize the results obtained with the microchannels nos.2, 3 and 4 is shown in Fig. 2.20. The dimensionless flow rate $1/\dot{m}^*$, little sensitive to the aspect ratio a^* , is plotted in function of $Kn'_{0,M}$, for an inlet over outlet pressure ratio $\Pi=1.8$. Up to $Kn'_{0,M}=0.05$, the first-order (NS1) and the second-order (NS2) slip flow models predict the same flow rate. For higher values of Kn'_0 , the deviation is significant and the experimental data are in very good agreement with the NS2 model, up to $Kn'_{0,M}=0.25$. Beyond that, the agreement is not so fine although the experimental data are closer to NS2 than to NS1 predictions.

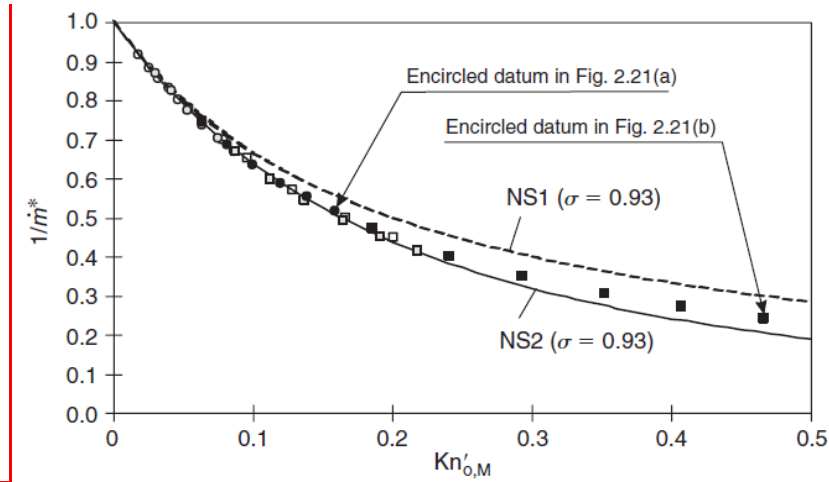


Fig. 2.20. Inverse reduced flow rate ($1/\dot{m}^*$) in rectangular microchannels: comparison of experimental data with NS1 and NS2 slip flow models; $\Pi=1.8$; $T=294.2$ K; microchannels no.2 (white), no.3 (grey), no.4 (black); gas: N₂ (circle) and He (square).

The experiments of Maurer *et al.* (2003) follow the same trend. The latest precise experimental data provided in the literature (Table 2.6) concern different gases and are relative to microchannels with cross-sections of comparable lengths l and depths $2h$, but manufactured with different processes. Arkilic *et al.* (2001) used two silicon wafers facing each other, Colin *et al.* (2004) studied microchannels etched by DRIE in silicon and covered with a glass sealed by anodic bonding, whereas Maurer *et al.* (2003) used microchannels etched in glass and covered with silicon. The lower and upper walls of the microchannels fabricated by Zohar *et al.* (2002) were obtained by deposition of low-stress silicon nitride layers on a silicon wafer.

Table 2.6

Recent experimental data of gas flows in rectangular microchannels.

Reference	Microchannels dimensions			Experimental conditions		Theoretical comparison	
	$l(\mu\text{m})$	$2h(\mu\text{m})$	$a^*(\%)$	Gas	Kn'_0	σ	Model
Shih <i>et al.</i> (1996)	4000	1.2	3.0	He	0.16	1.16	Plane, NS1
				N ₂	0.055	0.99	
Arkilic <i>et al.</i> (2001)	7490	1.33	2.5	Ar	0.05–0.41	0.80	Plane, NS1
				N ₂	0.05–0.34	0.80	
				CO ₂	0.03–0.44	0.80	
Lalonde (2001); Colin <i>et al.</i> (2004)	5000	0.54–4.48	1.1–8.7	N ₂	0.004–0.16	0.93; 1.00	Rectangular, NS2, $k_2 = k_{2,M}$, $A_2 = -9/8$; plane, QHD1, $k_2 = k_{2,VSS}$
				He	0.029–0.47	0.93; 1.00	
Zohar <i>et al.</i> (2002)	4000	0.53; 0.97	1.3; 2.4	He	0.21; 0.38	1.00	Plane, NS1, $k_2 = k_{2,M}$
				Ar	0.11; 0.20	1.00	
				N ₂	0.065; 0.12	1.00	
Maurer <i>et al.</i> (2003)	10,000	1.14	0.6	N ₂	0.054–1.1	0.87	Plane, NS2, $A_2 = -0.23_{(N_2)}$,
				He	0.17–1.46	0.91	$A_2 = -0.26_{(He)}$

A smart comparison of the data provided by these authors is also tricky, since the aspect ratios a^* of the sections are different from one study to another. The two latter studies (Maurer *et al.*, 2003; Colin *et al.*, 2004) confirm that an adequate slip flow model based on second-order boundary conditions can be precise for high Knudsen numbers that usually apply to the transition regime.

The experimental data from Lalonde have also been compared (Colin *et al.*, 2003) to the mass flow predicted by a QHD model with first-order boundary conditions (QHD1). For the microchannel no.4, with a low aspect ratio $a^*=0.011$, the plane assumption is acceptable and Eq. (2.111) is valid. Figure 2.21 shows that the QHD1 model could be accurate for Knudsen numbers higher than the NS2 model, but with a different accommodation coefficient, $\sigma = 1$, both for nitrogen and helium. Actually, the analysis of this result should be qualified. For this comparison, the mean free path was calculated using the formula given by Maxwell ($k_{2,M} = \sqrt{\pi/2}$) for the NS2 as well as for the QHD1 simulations, but the Schmidt value (0.74 for nitrogen and 0.77 for helium) used for the QHD1 simulation is very close to the one obtained from a VSS model, that is given by Eq. (2.46). Another comparison of the QHD1 model with the data of Lalonde is proposed in Elizarova and Sheretov (2003), using $k_{2,VSS}$ calculated from the VSS model (*cf.* Table 2.1). In this case, the better fit is found for $\sigma = 1$ for helium and $\sigma = 0.93$ for nitrogen.

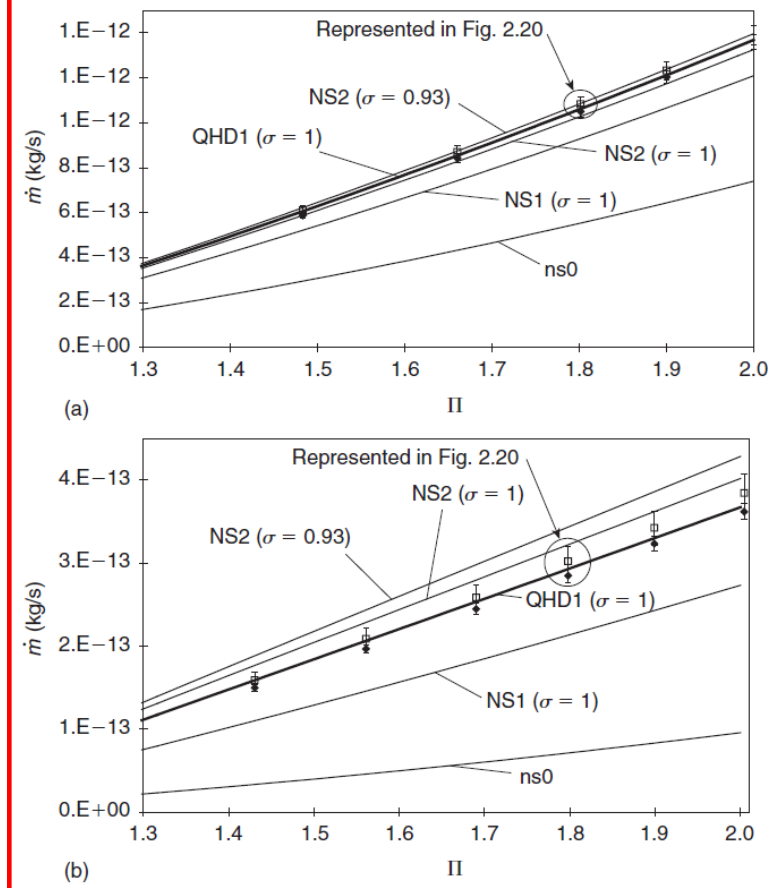


Fig. 2.21. Experimental data from Lalonde (outlet flow rate $\square$ and inlet flow rate $\blacklozenge$), compared to NS2 and QHD1 models. Microchannel no.4. $P_o = 75$ kPa, $T = 294.2$ K. (a) Gas: N_2 ; $Sc=0.74$; the encircled data correspond to $Kn'_{0,M}=0.15$. (b): Gas: He; $Sc=0.77$; the encircled data correspond to $Kn'_{0,M}=0.47$.

To conclude, it is clear that the NS2 or QHD1 models improve the accuracy of the NS1 model, but further studies that include smart experimental data are necessary to dissociate the roles of the accommodation coefficient, the collision model and the slip flow model.

2.3.5.3. Pressure data

Pressure measurements along a microchannel have been successfully performed by different groups (Liu et al., 1995; Lee et al., 2002; Zohar et al., 2002; Jang and Wereley, 2004), using integrated pressure microsensors connected to the microchannels via capillary connections (Fig. 2.22). These experiments confirm the theoretical pressure distribution shown in Fig. 2.11.

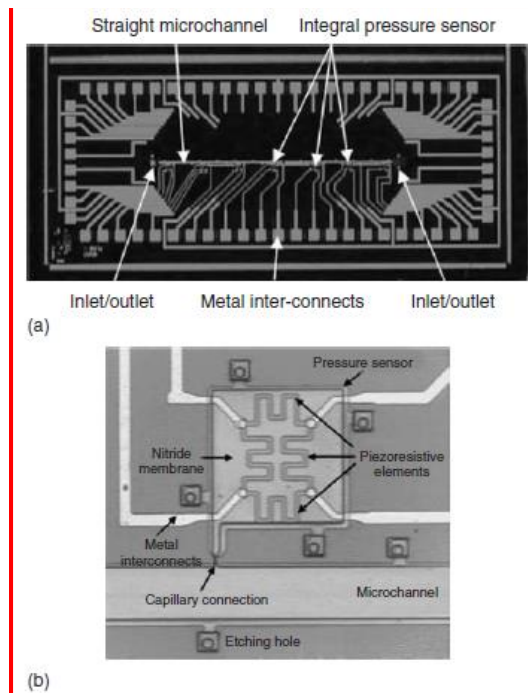


Fig. 2.22. Microchannel with integrated pressure sensors (Zohar et al., 2002).

(a) Microchannel with inlet and outlet;

(b) close-up of the pressure sensor and its capillary connection with the microchannel.

2.3.5.4. Flow visualization

Although micro-particle image velocimetry (micro-PIV) has been widely used to visualize liquid microflows for several years (Wereley et al., 1998), this technique has not been successfully applied to gaseous microflows as yet due to seeding and inertia issues (Wereley et al., 2002; Sinton, 2004). The molecular tagging velocimetry (MTV) could be an alternative and interesting technique for gas microflows visualization. This method was used to obtain the velocity profile of water in microtubes 705 μm in diameter, with a spatial resolution lower than 4% (Webb and Maynes, 1999). Recently, the technique was adapted to the analysis of gaseous subsonic or supersonic microjets generated by a 1 mm diameter nozzle. The gas was nitrogen, seeded with acetone and the spatial resolution was around 10 μm , with a maximal uncertainty of 5% for a Knudsen number between 0.002 and 0.035, that is in the slip flow regime.

2.3.6. Entrance effects

In the previous sections, the microchannels have been considered as long microchannels and the entrance effects have been neglected.

If the microchannel is short, such that the hydrodynamic development length L_{hd} is no longer negligible compared to the microchannel length, entrance effects have to be taken into account by a numerical approach.

Rarefaction increases L_{hd} in the slip flow regime, due to slip at the walls. Barber and Emerson (2002) used a two-dimensional finite-volume Navier–Stokes solver to simulate the flow of a gas entering a plane microchannel in the slip- flow regime, with first-order boundary conditions. Their simulated data, for $Kn \in [0; 0.1]$ and $Re \in [0; 400]$ were fitted by a least-squares technique and led to the following expression for the hydrodynamic development length:

$$\frac{L_{hd}}{D_H} = \frac{0.332}{0.0271 Re + 1} + 0.011 Re \frac{1 + 14.78 Kn}{1 + 9.78 Kn} \quad (2.160)$$

which generalizes the expression

$$L_{hd} / D_H = 0.315 / (0.0175 Re + 1) + 0.011 Re$$

found by Chen (1973) for a no-slip flow with a slight modification of the non-rarefied part of the equation.

In Eq. (2.160), the Knudsen number $Kn = \lambda / D_H$ and the Reynolds number

$Re = \rho u_z D_H / \mu$ are based on the hydraulic diameter $D_H = 4h$ of the microchannel, $2h$ in depth. The hydrodynamic development length is arbitrarily defined as the longitudinal distance required for the centerline velocity to reach 99% of its fully developed value. Equation (2.160) shows that for $Kn = 0.1$ and $Re = 400$, slip at the wall increases the entrance length of about 25%.